

School Learning Platform

Burak, Sebastian, Yassa

Agenda

Technology Stack

Implementation

- Burak - User Account Module
- Sebastian - Lecture Module
- Yassa - Learning Module

Demo

Technology Stack

Frontend	Backend	Database
React 	Node 	Mongo 
Vite 	Express 	Mongoose 
Tailwind 		Vercel Blob 
		

Project Description - Burak

User account system with the following features:

- Registration: Users can register using their name, email, and password
- Login: Users can log in using their email and password
- Password Recovery: Users can recover their password through email links
- Security: Implement bot and automation prevention mechanisms
- Roles: There are two roles - student and teacher
- Data Requirements: Find a suitable dataset to populate the database with 500 users.

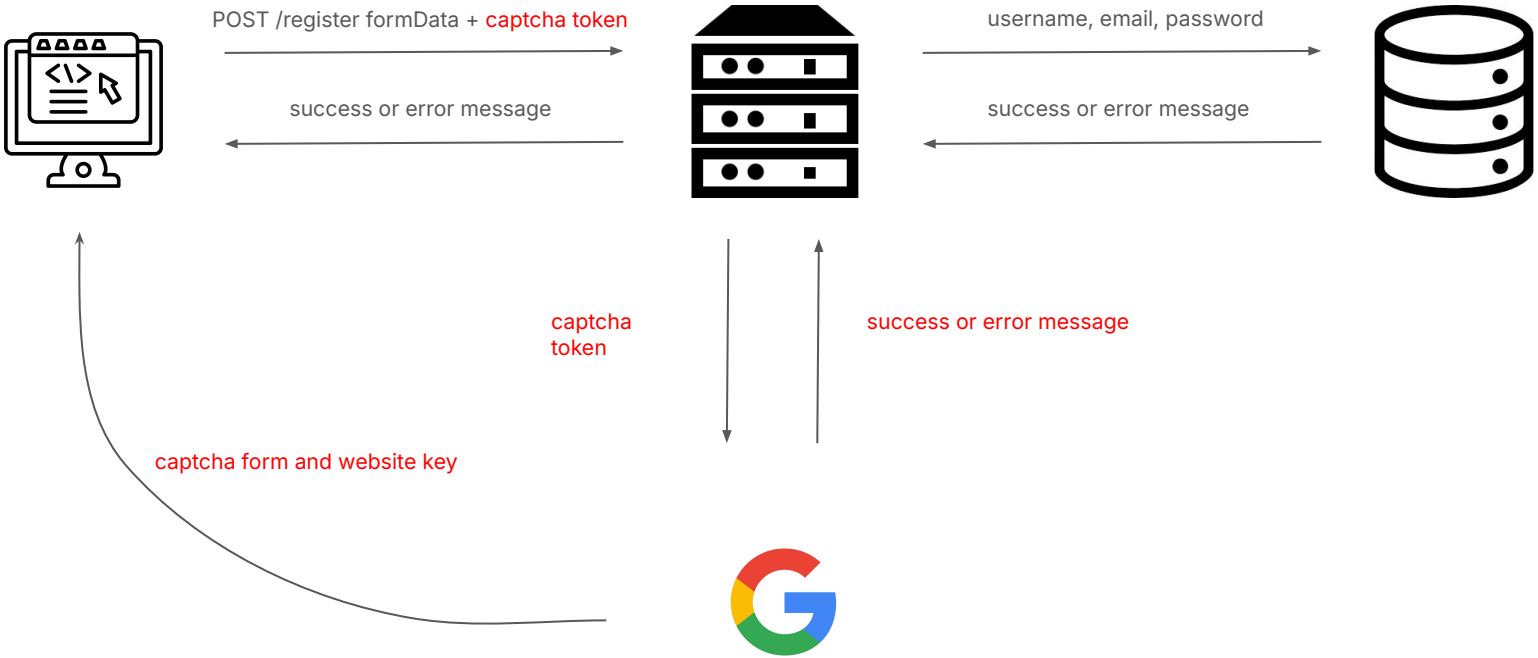
Implementation - Register and Login



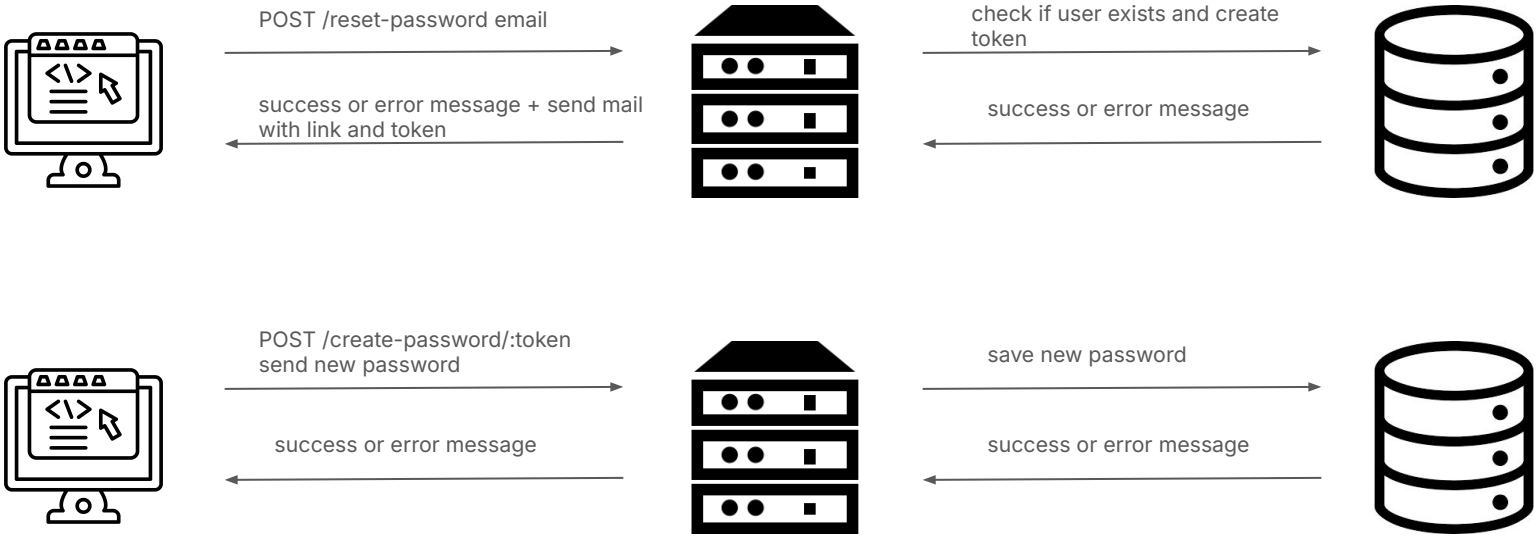
Implementation - Changing User Roles



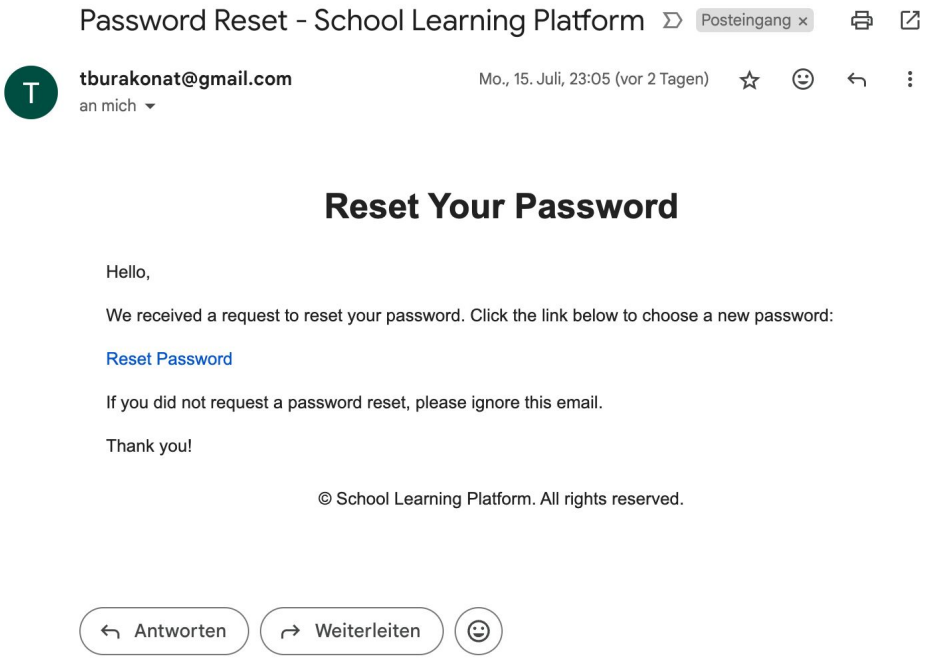
Implementation - Captchas

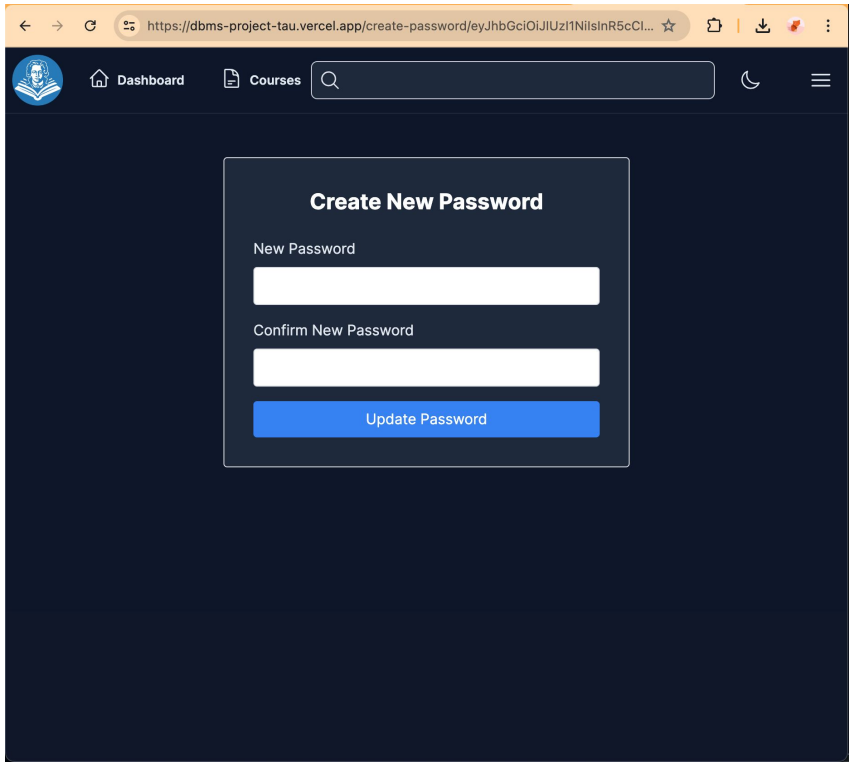


Implementation - Password Reset



Implementation - Password Reset





Implementation - User Roles



Student role is the default for new users

-



Teacher role can be set by admin user in the app

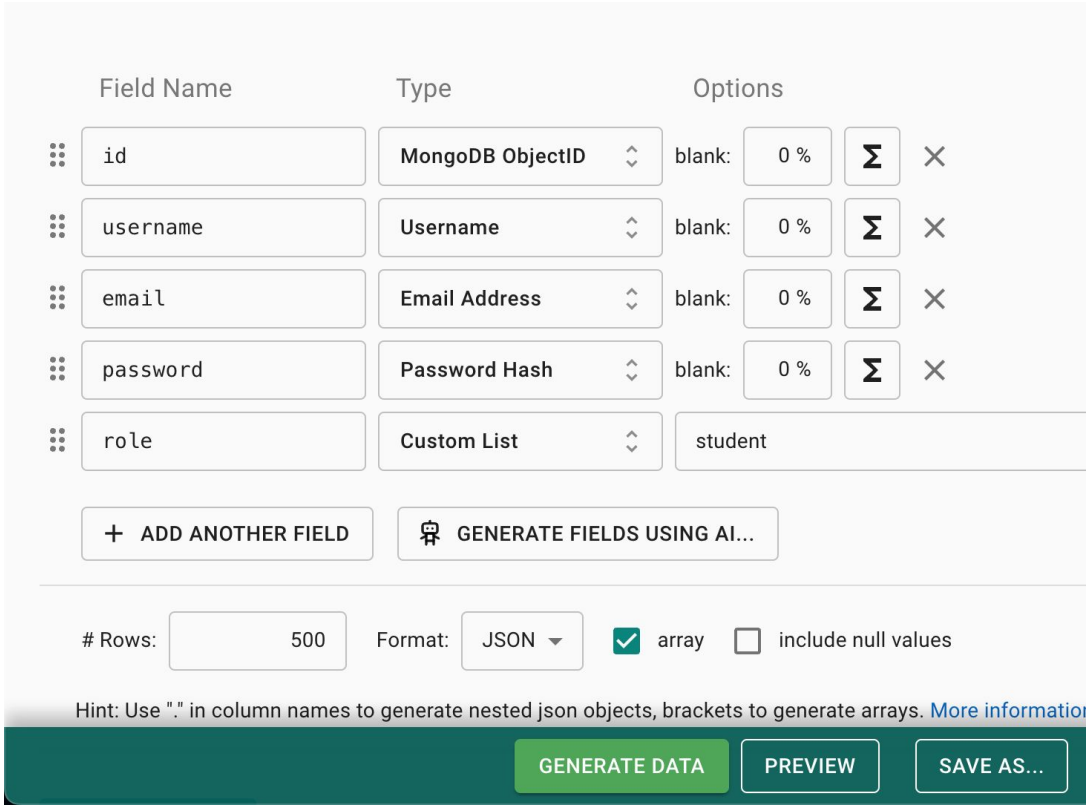
see students of your courses



Admin role can be set by admin user in the app

see all users and change their roles

Implementation - Data Requirements




```
[
    {
        "_id": { "$oid": "6676b64efc13ae49152347b0" },
        "username": "ltomankowski0",
        "email": "efarington0@tinyurl.com",
        "password": "$2a$04$9mG0bAV1uQztRk0/b/Z3r.
        brMHSG5HgrYoGVGWdWo4s2DBnGyQfau",
        "role": "student"
    },
    {
        "_id": { "$oid": "6676b64efc13ae49152347b1" },
        "username": "tsaunt1",
        "email": "kbeartup1@shop-pro.jp",
        "password": "$2a$04$hAVgd6HRBAtbkgs3LSV55.6PLIkMleoIThPaonV.yBsyDA
        yf1ZS",
        "role": "student"
    },
    {
        "_id": { "$oid": "6676b64efc13ae49152347b2" },
        "username": "wkidson2",
        "email": "dambroziak2@desdev.cn",
        "password": "$2a$04$DGxr9ya/JxHA5Daz2IqKdOciIQs53BCu8jZ.
        sv3bMpFNczjbJE7IS",
        "role": "student"
    },
```

Project Description - Sebastian

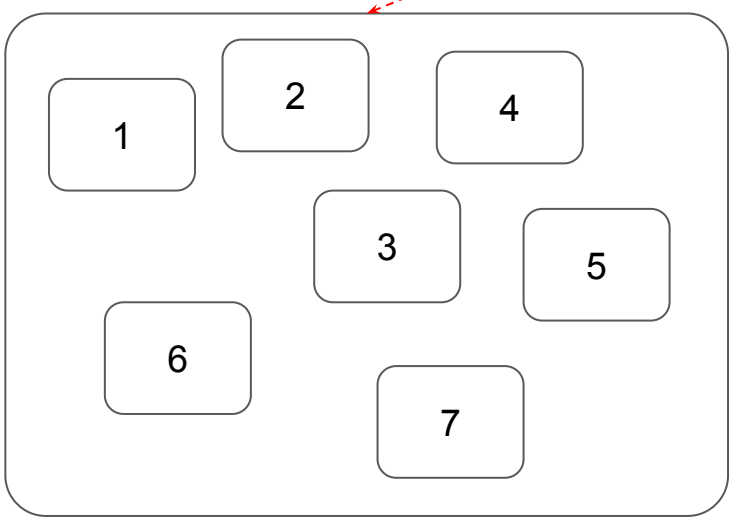
Lecture Module with the following features:

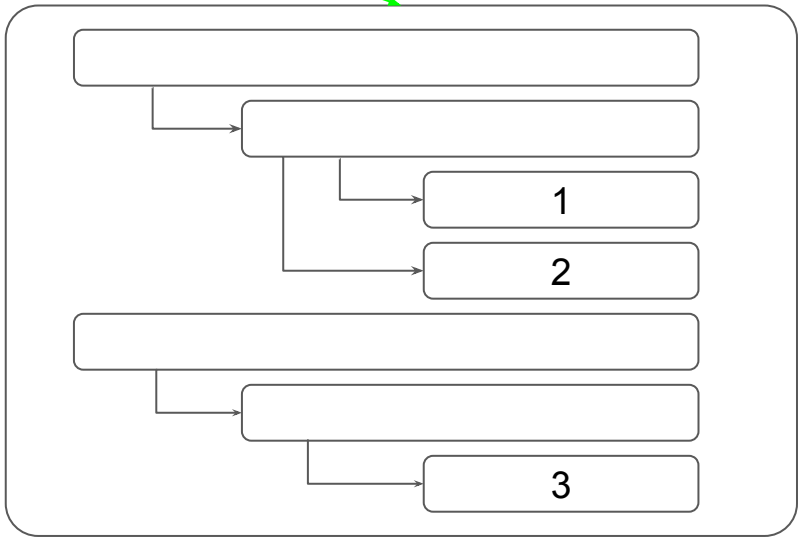
- Lecture List: The page loads a list of lectures with their respective dates
- Lecture Content: Each lecture displays a topic with a video, text and images
- Interactive Elements: Students can write comments, highlight text and add notes
- Data Persistence: Notes and questions are saved and loaded upon return
- Admin Page: Pages for creating content on the site

Implementation - No lecture list for all lectures

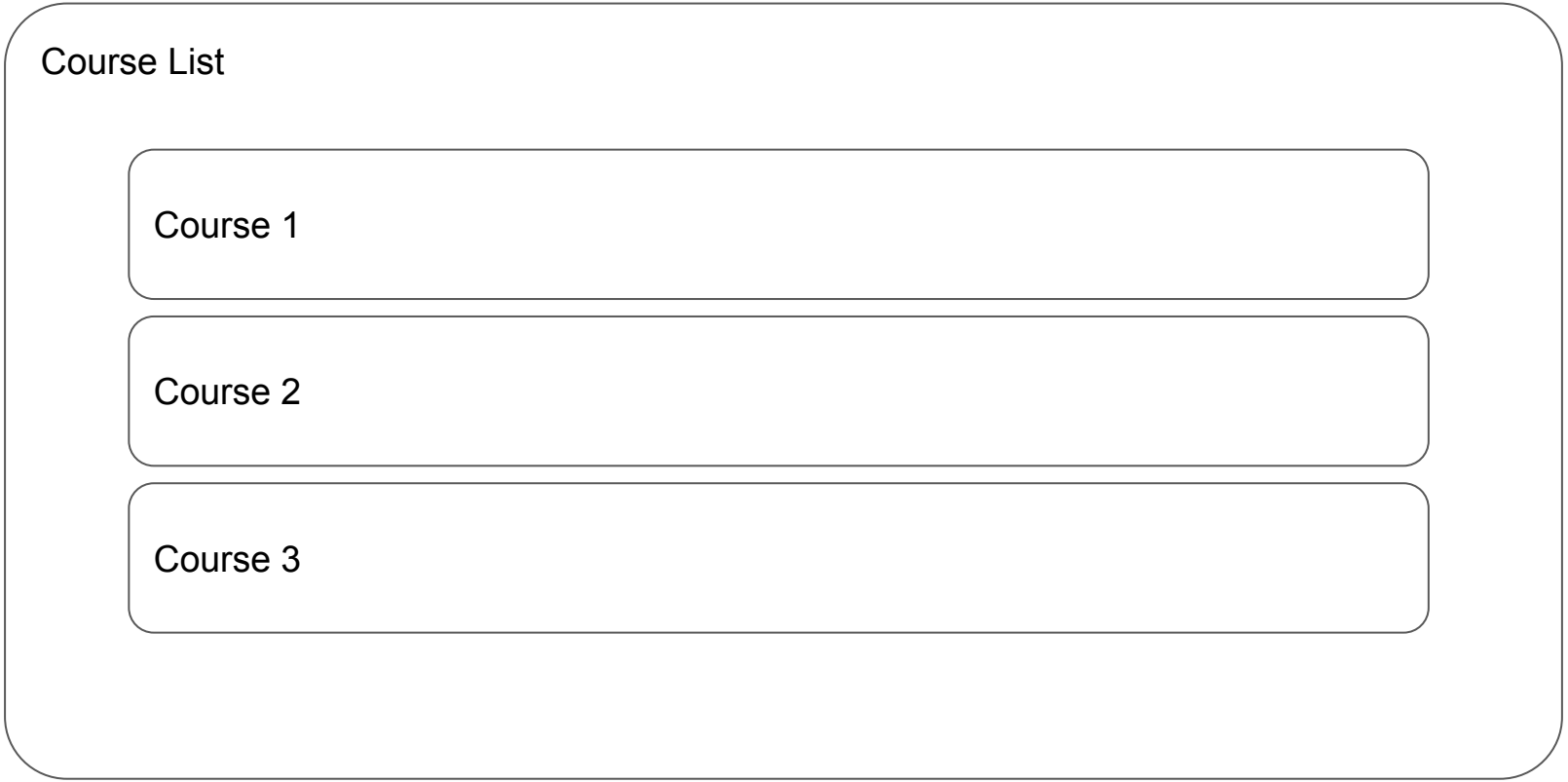
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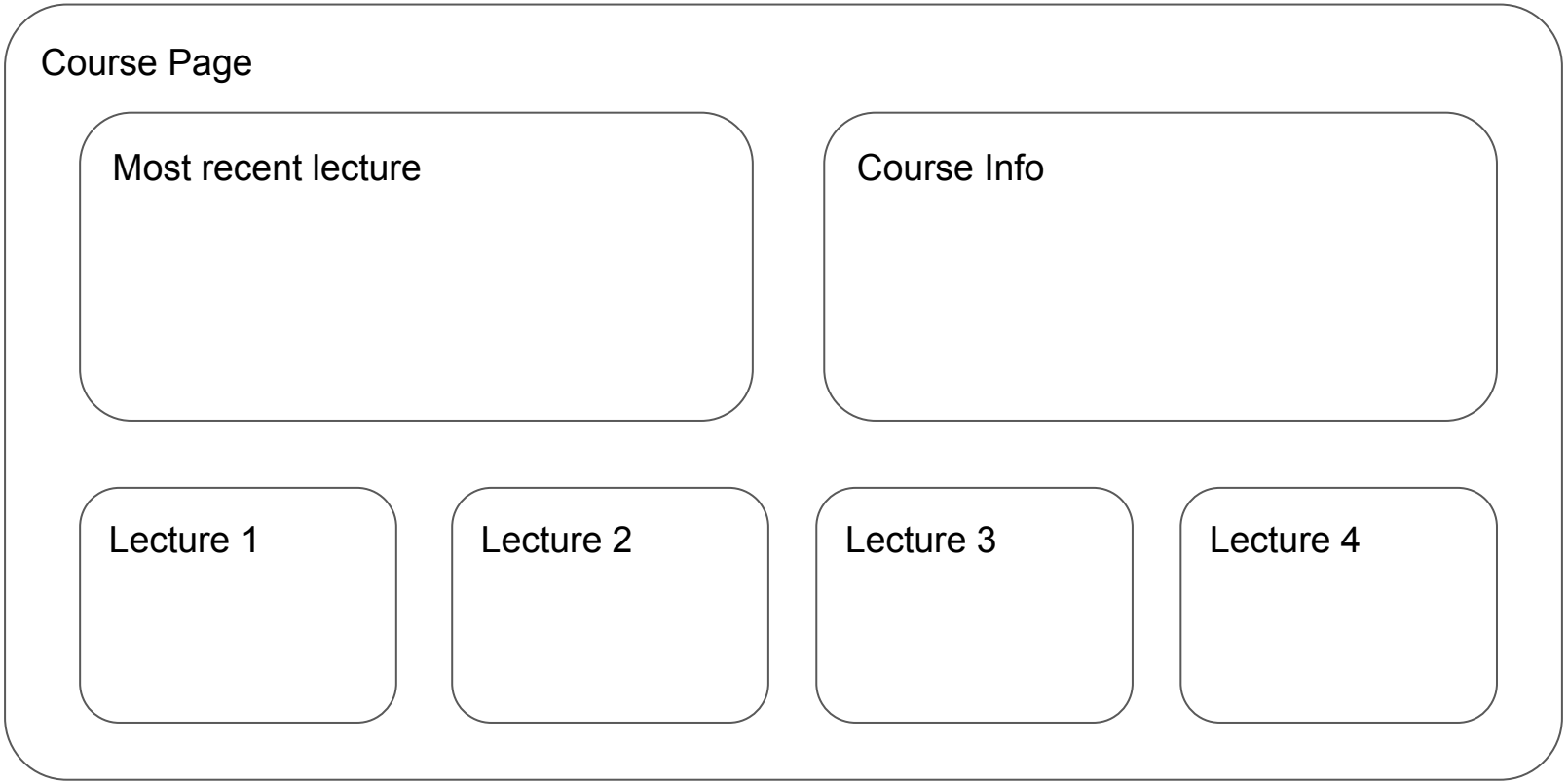




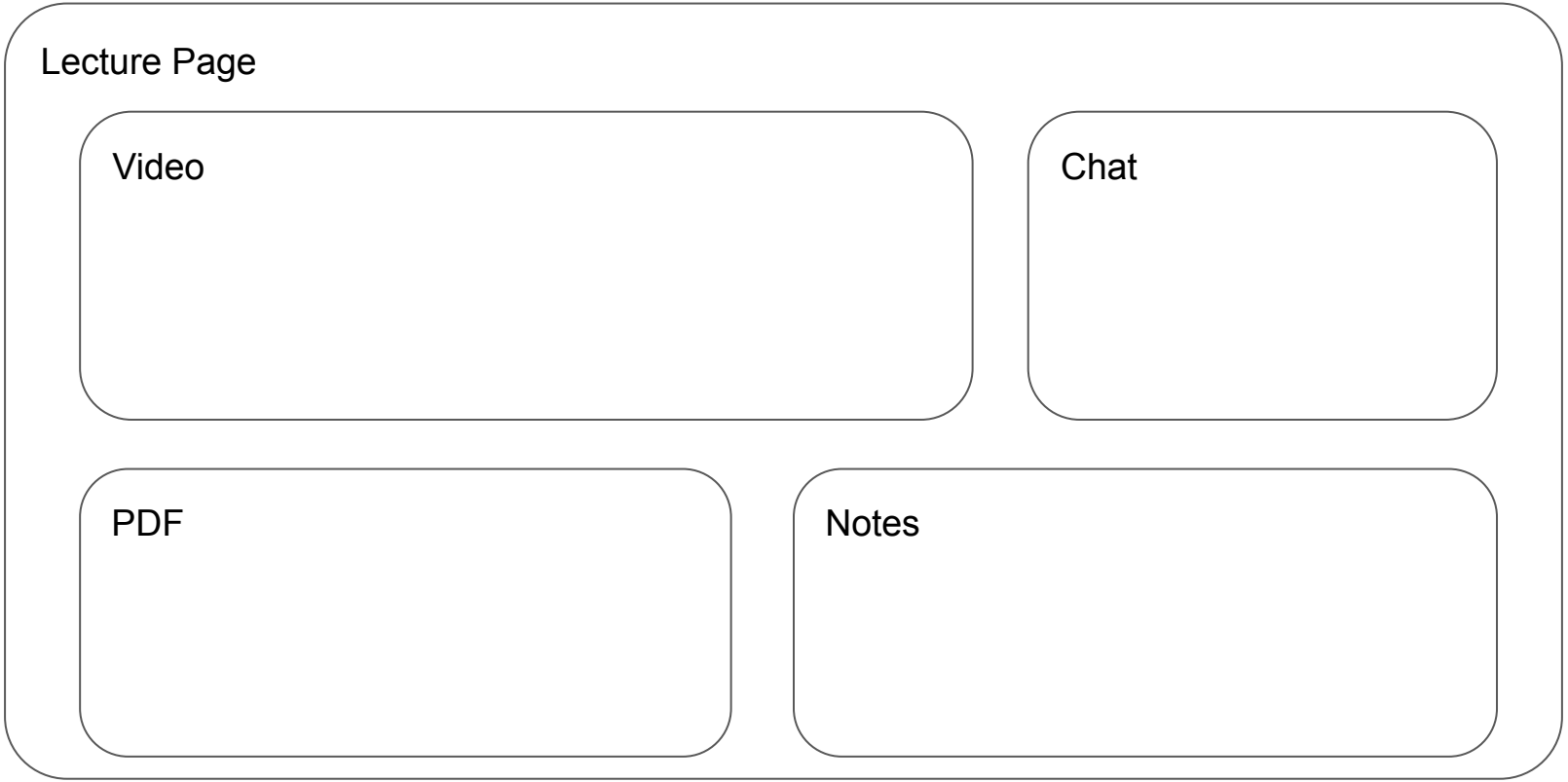
Implementation - Course List



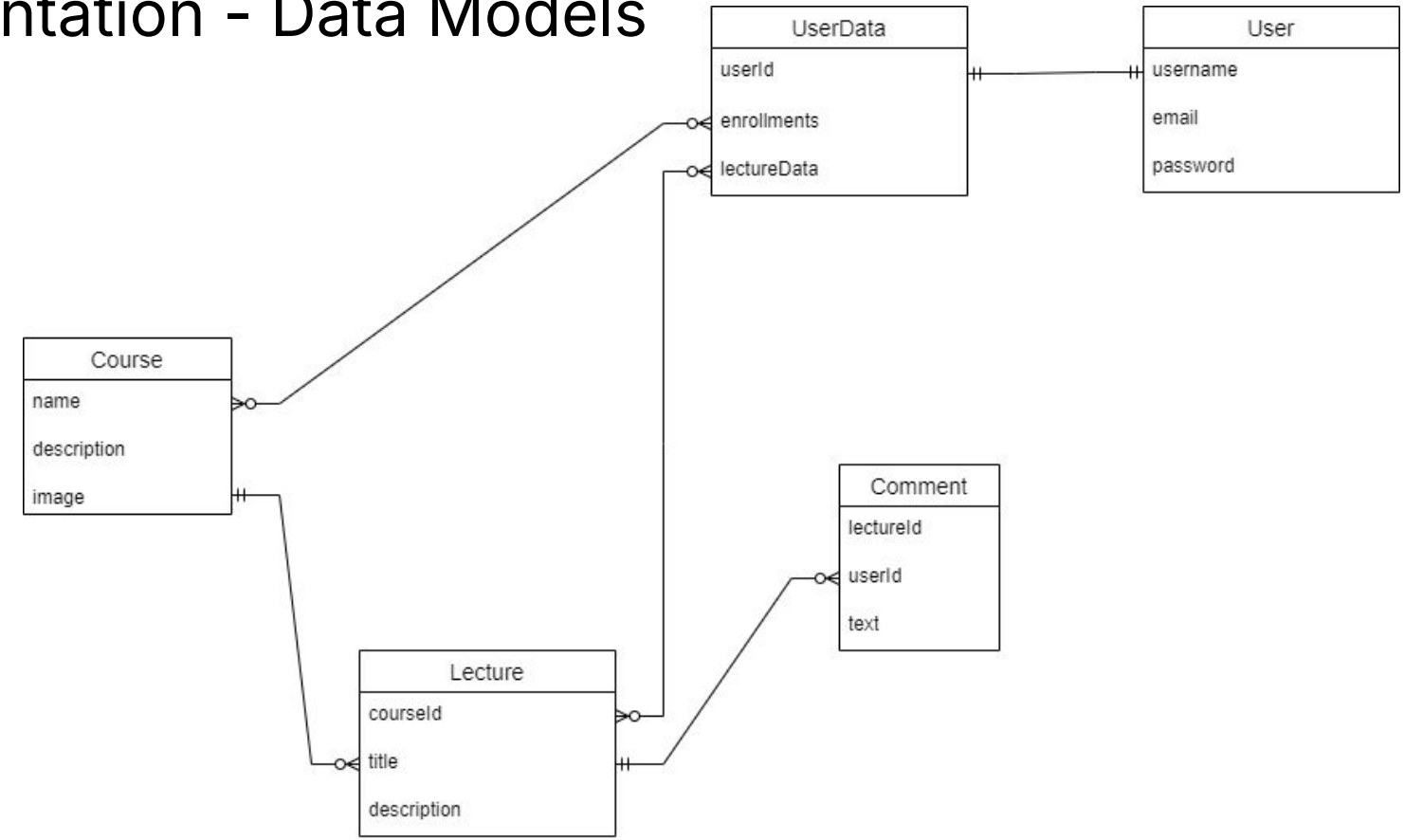
Implementation - Lecture List



Implementation - Lecture Content



Implementation - Data Models



Project Description - Yassamin

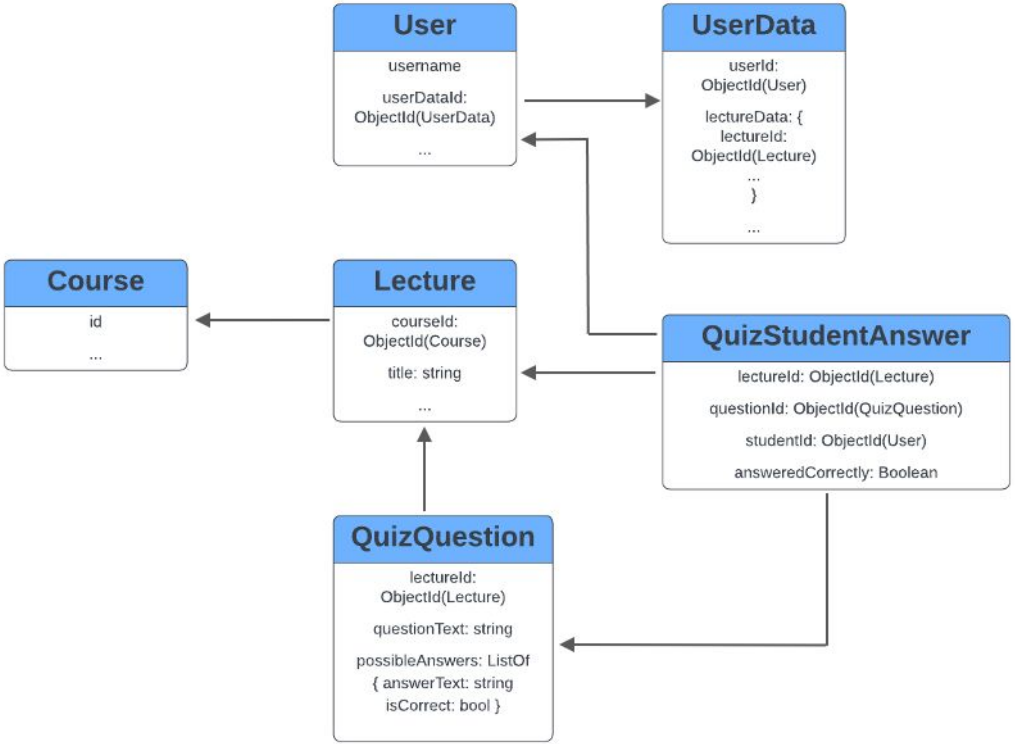
Learning Module with the following features:

- **Multiple-choice questions** categorized by lectures
- **Students must have viewed related lecture** to access questions of the subject
- **Quiz performance is tracked** and if this falls below 20%, receives suggestion to watch lecture
- Questions are presented using **Spaced Repetition**

Considerations

- **How should these questions/answers be created and managed**? (Teachers)
- What should the **“quiz-taking” experience** be? (Students)
- **How can we model the data** such that we can keep track of performance & know which question to ask next?
  - **Performance**: we must save both correct & incorrect answers per question
  - **Spaced repetition**: we must know ***when*** the correct and incorrect answers were submitted

Data model


User
username
userDataId: ObjectId(UserData)
...
UserData
userId: ObjectId(User)
lectureData: { lectureId: ObjectId(Lecture) ... }
...
Course
id
...
Lecture
courseId: ObjectId(Course)
title: string
...
QuizStudentAnswer
lectureId: ObjectId(Lecture)
questionId: ObjectId(QuizQuestion)
studentId: ObjectId(User)
answeredCorrectly: Boolean
QuizQuestion
lectureId: ObjectId(Lecture)
questionText: string
possibleAnswers: ListOf { answerText: string isCorrect: bool }

Data model

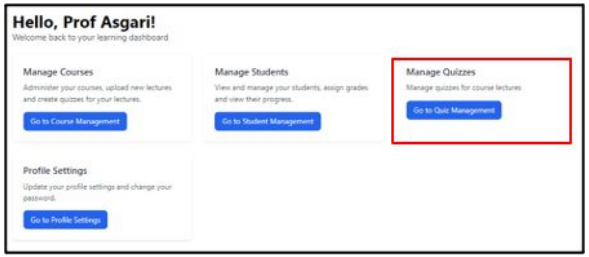
One Quiz-per-Lecture

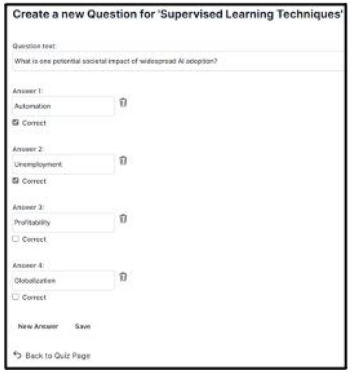
- → Each lecture can only have 1 quiz. If a student performs poorly, they are linked back to that lecture.
- → Requirements specified organizing Quiz by “subject”, but each lecture covering 1 subject seems like a reasonable assumption and doing stuff like e.g. tags would be complicated.

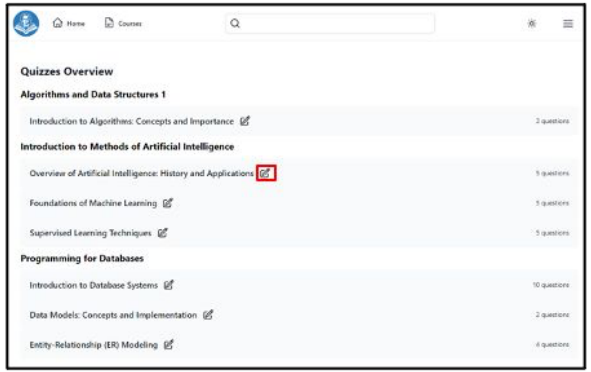
No “Quiz” model → this is a “front-end” (or “feature”) consideration

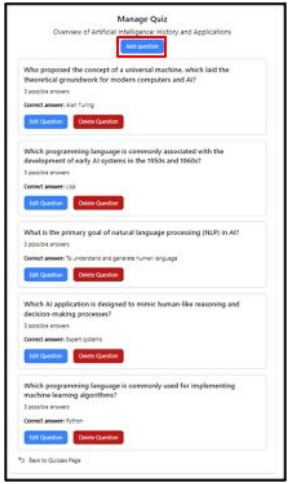
- → A quiz consists of questions associated with a lecture and their respective answers.
- → Student quiz results can be calculated on-demand by reviewing their answers.

Teacher UX (1)

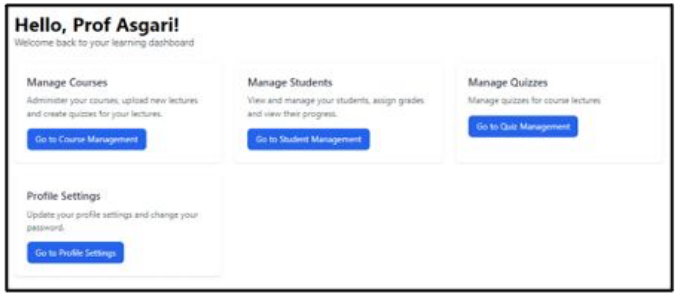

Hello, Prof Asgari!
Welcome back to your learning dashboard
Manage Courses
Administer your courses, upload new lectures and create quizzes for your lectures.
Go to Course Management
Manage Students
View and manage your students, assign grades and view their progress.
Go to Student Management
Manage Quizzes
Manage quizzes for course lectures
Go to Quiz Management
Profile Settings
Update your profile settings and change your password.
Go to Profile Settings

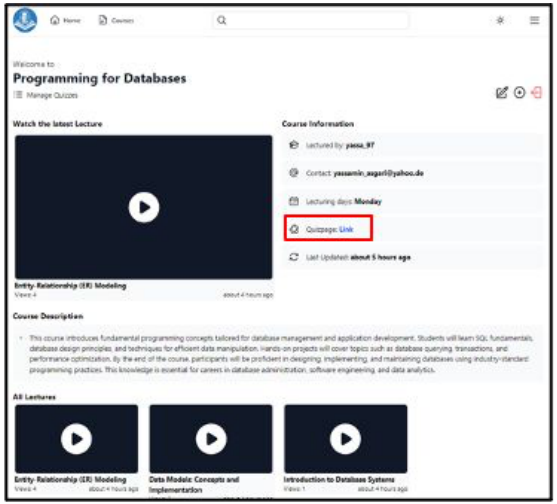

Create a new Question for 'Supervised Learning Techniques'
Question text:
What is one potential societal impact of widespread AI adoption?
Answer 1:
Automation
Correct
Answer 2:
Unemployment
Correct
Answer 3:
Profitability
Correct
Answer 4:
Globalization
Correct
New Answer
Save
Back to Quiz Page


Home
Courses
Quizzes Overview
Algorithms and Data Structures 1
Introduction to Algorithms: Concepts and Importance
2 questions
Introduction to Methods of Artificial Intelligence
Overview of Artificial Intelligence: History and Applications
5 questions
Foundations of Machine Learning
5 questions
Supervised Learning Techniques
5 questions
Programming for Databases
Introduction to Database Systems
10 questions
Data Models: Concepts and Implementation
2 questions
Entity-Relationship (ER) Modeling
4 questions


Manage Quiz
Overview of Artificial Intelligence: History and Applications
Add Question
Who proposed the concept of a universal machine, which laid the theoretical groundwork for modern computers and AI?
3 possible answers
Correct answer: Alan Turing
Edit Question
Delete Question
Which programming language is commonly associated with the development of early AI systems in the 1950s and 1960s?
3 possible answers
Correct answer: Lisp
Edit Question
Delete Question
What is the primary goal of natural language processing (NLP) in AI?
3 possible answers
Correct answer: To understand and generate human language
Edit Question
Delete Question
Which AI application is designed to mimic human-like reasoning and decision-making processes?
3 possible answers
Correct answer: Expert systems
Edit Question
Delete Question
Which programming language is commonly used for implementing machine learning algorithms?
3 possible answers
Correct answer: Python
Edit Question
Delete Question
Back to Quizzes Page

Teacher UX (2)







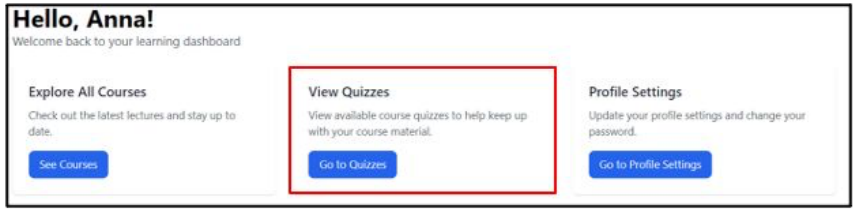
...See previous slide

Spaced repetition

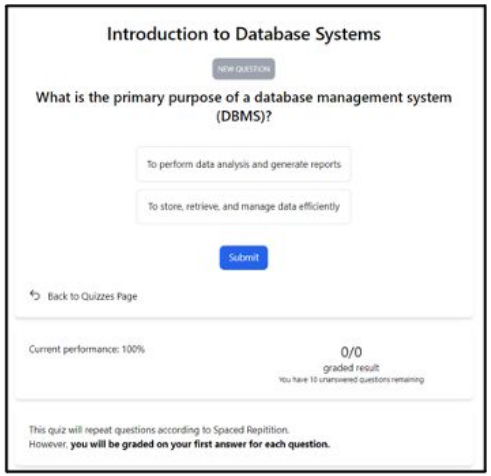
- Main idea: Reinforce knowledge just before it is likely to be forgotten.
- Is a learning technique that involves active recall (reviewing), where correctly answered questions are shown less frequently.
- Runs indefinitely.
- Recorded all student answers with timestamps.

1. track the timing of their last correct or incorrect response
2. focus is on identifying the most recent answer among all questions.

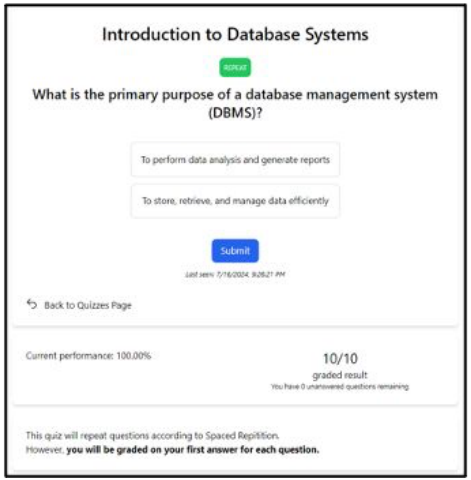
Student UX (1)


Hello, Anna!
Welcome back to your learning dashboard
Explore All Courses
Check out the latest lectures and stay up to date.
See Courses
View Quizzes
View available course quizzes to help keep up with your course material.
Go to Quizzes
Profile Settings
Update your profile settings and change your password.
Go to Profile Settings


Quizzes Overview
Programming for Databases
Introduction to Database Systems
10 questions
Data Models: Concepts and Implementation
2 questions
Entity-Relationship (ER) Modeling
4 questions


Introduction to Database Systems
NEW QUESTION
What is the primary purpose of a database management system (DBMS)?
To perform data analysis and generate reports
To store, retrieve, and manage data efficiently
Submit
Back to Quizzes Page
Current performance: 100%
0/0
graded result
You have 10 unanswered questions remaining
This quiz will repeat questions according to Spaced Repitition.
However, you will be graded on your first answer for each question.

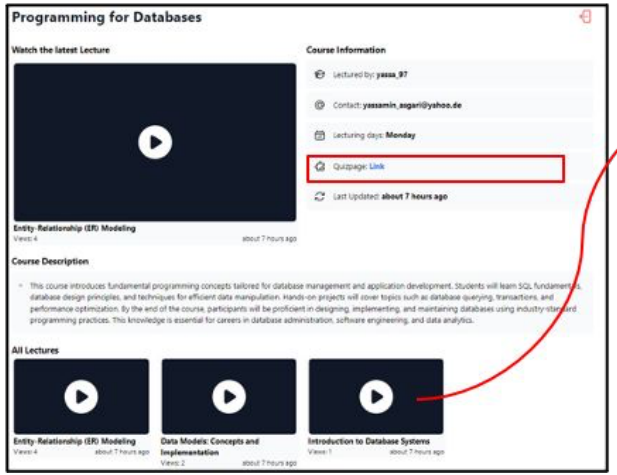

Introduction to Database Systems
REVIEW
Which of the following is an example of a relational database management system (RDBMS)?
MongoDB
MySQL
NeoAJ
Submit
Back to Quizzes Page
Current performance: 18.18%
Your performance has fallen below 20% of questions answered correctly. Perhaps you would like to review the lecture material?
2/10
graded result
You have 0 unanswered questions remaining

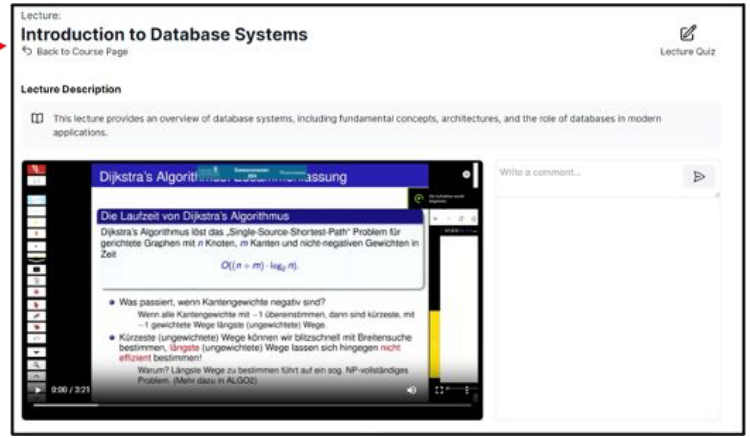

Introduction to Database Systems
REPEAT
What is the primary purpose of a database management system (DBMS)?
To perform data analysis and generate reports
To store, retrieve, and manage data efficiently
Submit
Back to Quizzes Page
Current performance: 100.00%
10/10
graded result
You have 0 unanswered questions remaining
This quiz will repeat questions according to Spaced Repitition.
However, you will be graded on your first answer for each question.

Student UX (2)

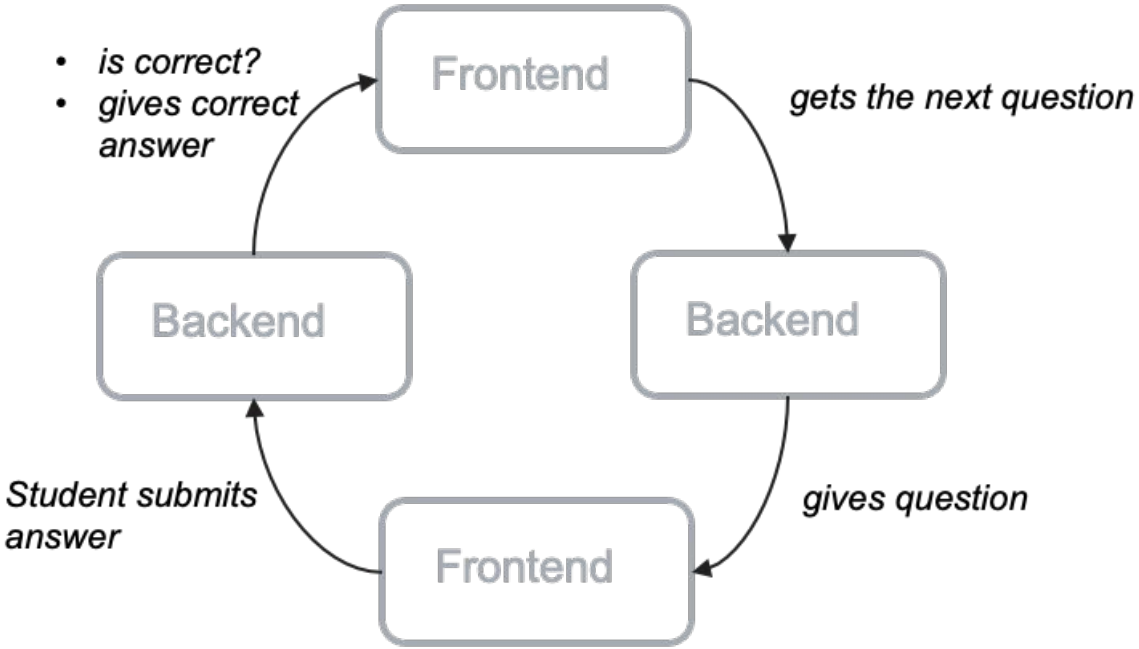

Hello, Anna!
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Explore All Courses
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See Courses
View Quizzes
View available course quizzes to help keep up with your course material.
Go to Quizzes
Profile Settings
Update your profile settings and change your password.
Go to Profile Settings


Available Courses
Find a course you are interested in
Algorithms and Data Structures 1
Updated 7/16/2024, 2:21:49 PM
Introduction to Methods of Artificial Intelligence
Updated 7/16/2024, 2:23:06 PM
Programming for Databases
Updated 7/16/2024, 2:24:08 PM


Programming for Databases
Watch the latest Lecture
Course Information
Lectured by: yassa_97
Contact: yassamin_asgari@yahoo.de
Lecturing days: Monday
Quizpage: Link
Last Updated: about 7 hours ago
Entity-Relationship (ER) Modeling
Course Description
All Lectures
Entity-Relationship (ER) Modeling
Data Models: Concepts and Implementation
Introduction to Database Systems


Lecture:
Introduction to Database Systems
Back to Course Page
Lecture Quiz
Lecture Description
This lecture provides an overview of database systems, including fundamental concepts, architectures, and the role of databases in modern applications.
Write a comment...

The “Quiz-Taking” Loop



Possible improvements

- **What if we wanted to supported multiple quizzes per lecture?**
  -TODO: Add a Quiz model for that?

- **Is it wasteful to have to calculate the student's Quiz results each time?**
  -Probably could have a Collection for that as an aggregate

- **Implement different design choices**
  -Multiple-Choice question (current performance, graded result, overview, etc.)